

HOBURG FYR, Sweden

WMO: **026790**

Lat: **56.9209N** Lon: **18.1506E** Elev: **39** StdP: **100.86** Time Zone: **1.00 (EUC)** Period: **94-19** WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF	
				99.6%			99%			0.4%		1%					
		99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	2	-7.0	-5.4	-13.2	1.2	-4.4	-11.1	1.5	-3.9	15.8	1.3	14.6	1.7	5.0	20	0.708	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB		
			0.4%		1%		2%		0.4%		1%		2%				
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	8	4.8	24.2	18.3	22.8	17.6	21.5	17.2	20.0	22.3	19.0	21.4	18.1	20.4	4.7	140	(2)

	Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
	0.4%			1%			2%			0.4%		1%		2%			
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	19.1	14.0	21.1	18.1	13.0	20.2	17.1	12.3	19.5	57.3	22.4	54.1	21.5	51.4	20.5	24.3	(3)

Extreme Annual Design Conditions

	Extreme Annual WS				Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
					Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
	1%	2.5%	5%		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	13.8	12.3	11.1	DB	-8.3	26.1	2.5	2.2	-10.1	27.7	-11.6	29.0	-13.1	30.3	-14.9	31.9	(4)
(5)				WB	-9.4	20.7	2.2	1.6	-11.0	21.8	-12.3	22.7	-13.5	23.6	-15.1	24.7	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours		DBAvg	8.2	0.6	0.2	1.6	5.2	9.7	14.1	17.6	17.9	14.0	9.1	5.2	2.4	(6)
(7)			DBStd	6.80	2.85	2.98	2.63	2.68	2.92	2.30	2.52	2.26	2.38	2.65	2.62	2.95	(7)
(8)			HDD10.0	1444	292	274	261	147	41	1	0	0	1	48	144	236	(8)
(9)			HDD18.3	3756	550	508	519	394	269	128	46	36	132	286	394	494	(9)
(10)			CDD10.0	778	0	0	0	2	31	124	234	244	121	21	1	0	(10)
(11)			CDD18.3	44	0	0	0	0	0	1	22	21	1	0	0	0	(11)
(12)			CDH23.3	86	0	0	0	0	1	1	56	28	1	0	0	0	(12)
(13)			CDH26.7	4	0	0	0	0	0	0	3	1	0	0	0	0	(13)
(14)	Wind		WSAvg	5.8	6.8	6.3	5.9	5.4	5.0	5.1	4.9	5.1	5.8	6.3	6.4	7.0	(14)
(15)	Precipitation		PrecAvg	507	41	32	31	28	34	36	47	52	56	47	55	49	(15)
(16)			PrecMax	711	122	77	75	84	111	97	145	117	130	127	109	88	(16)
(17)			PrecMin	376	6	8	2	2	7	0	1	6	4	6	13	13	(17)
(18)			PrecStd	84	26	14	17	17	24	24	36	28	31	28	21	23	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB		6.5	5.9	8.9	16.4	20.6	22.6	26.7	25.7	21.7	15.3	11.2	8.2	(19)
(20)			MCWB		5.6	4.3	5.3	9.9	14.3	15.3	19.3	19.3	16.6	13.5	10.3	7.4	(20)
(21)		2%	DB		5.6	5.0	6.7	13.1	18.0	20.9	24.9	23.9	19.5	14.3	10.1	7.5	(21)
(22)			MCWB		4.9	4.0	4.6	7.9	12.7	15.2	19.0	18.1	16.0	12.8	9.3	6.8	(22)
(23)		5%	DB		4.9	4.4	5.6	10.9	15.9	19.3	23.3	22.7	18.3	13.4	9.2	6.8	(23)
(24)			MCWB		4.2	3.5	4.0	7.0	11.6	14.7	18.1	17.6	15.4	12.1	8.5	6.2	(24)
(25)		10%	DB		4.2	3.8	4.7	9.2	14.2	18.0	21.9	21.5	17.3	12.6	8.5	6.0	(25)
(26)			MCWB		3.6	3.0	3.5	6.3	10.7	14.2	17.6	17.4	14.8	11.5	7.8	5.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB		6.0	4.9	6.2	10.6	15.8	17.8	21.7	21.8	18.2	14.1	10.7	7.9	(27)
(28)			MCDB		6.2	5.3	7.9	15.5	19.3	20.7	24.0	24.2	20.1	14.8	10.9	8.0	(28)
(29)		2%	WB		5.2	4.3	5.1	8.7	13.8	16.5	20.2	19.9	17.0	13.2	9.5	7.0	(29)
(30)			MCDB		5.4	4.7	6.2	11.6	16.5	19.3	22.9	21.8	18.6	13.9	9.9	7.3	(30)
(31)		5%	WB		4.4	3.7	4.4	7.5	12.5	15.6	19.3	19.0	16.2	12.5	8.7	6.2	(31)
(32)			MCDB		4.7	4.1	5.3	10.2	15.0	18.1	22.0	21.1	17.7	13.2	9.1	6.6	(32)
(33)		10%	WB		3.7	3.2	3.7	6.6	11.3	14.8	18.4	18.2	15.4	11.7	8.0	5.4	(33)
(34)			MCDB		4.1	3.7	4.5	8.8	13.7	17.1	21.0	20.3	16.8	12.4	8.3	5.8	(34)
(35)	Mean Daily Temperature Range		MDBR		2.5	2.7	3.6	5.3	5.8	5.4	5.1	4.8	4.1	3.0	2.5	2.4	(35)
(36)		5% DB	MCDBR		2.5	2.9	4.6	8.2	8.4	7.5	7.3	6.2	4.9	3.3	2.8	2.3	(36)
(37)			MCWBR		2.6	2.7	3.3	4.4	4.8	3.7	3.4	3.4	3.1	2.8	2.9	2.6	(37)
(38)		5% WB	MCDBR		2.4	2.7	4.2	7.2	7.4	6.2	6.1	5.2	4.4	3.1	2.6	2.3	(38)
(39)			MCWBR		2.6	2.6	3.3	4.4	4.7	3.8	3.5	3.6	3.1	2.8	2.9	2.7	(39)
(40)	Clear-Sky Solar Irradiance		taub		0.305	0.332	0.348	0.382	0.377	0.368	0.400	0.392	0.362	0.350	0.334	0.298	(40)
(41)			taud		2.285	2.329	2.358	2.320	2.414	2.473	2.404	2.416	2.475	2.401	2.280	2.228	(41)
(42)			Ebn at Noon		573	716	808	831	862	874	834	815	789	687	521	464	(42)
(43)			Edh at Noon		51	73	91	109	106	101	107	99	81	68	51	41	(43)
(44)	All-Sky Solar Radiation		RadAvg		0.35	0.95	2.46	4.40	5.88	6.39	5.77	4.64	3.12	1.41	0.50	0.25	(44)
(45)			RadStd		0.06	0.17	0.23	0.47	0.44	0.33	0.51	0.40	0.34	0.22	0.08	0.04	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.39	N/A	+2.04	N/A	N/A	N/A	N/A	-130	+66	N/A	(46)
(47)	Regional (21 neighbors)	+0.36	N/A	+1.20	N/A	N/A	N/A	-101	-140	N/A	N/A	(47)

Nomenclature: See separate page